VISHAL NAGARAJAN

Embedded Software Engineer | Embedded C/C++ | Firmware | Microcontrollers
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SUMMARY

Electronics and Communication Engineering graduate with hands-on experience in embedded software development, microcontrollers, and hardware-software integration. Experienced in C/C++, Embedded C, CAN communication, and real-time embedded systems, with internship experience at Hindustan Aeronautics Limited involving MISRA C compliance and embedded software performance optimization. Interested in developing reliable firmware for automotive, industrial, and safety-critical systems.

WORK EXPERIENCE

HINDUSTAN AERONAUTICS LIMITED (HAL) *Bengaluru, Karnataka*
Intern *Jan 2024 – Feb 2024*

- Optimized embedded software algorithms, reducing processing latency by 1.04 seconds through performance-focused implementation.
- Applied MISRA C coding guidelines to existing embedded software, improving adherence to safety-critical coding practices.
- Collaborated with multidisciplinary teams during research and system-level work related to a Terrain Avoidance Warning System.
- Gained practical exposure to avionics, embedded software development, and safety-oriented engineering practices.

INDEPENDENT EMBEDDED SYSTEMS DEVELOPER *Coimbatore, TamilNadu*
Freelance/Self-employed *Feb 2024 - Present*

- Independently designed and developed 13 engineering prototypes, including 7 hardware and 6 software systems, across embedded systems, IoT, computer vision, and automation.
- Developed solutions using STM32, ESP32, Arduino, and Raspberry Pi, integrating camera, and embedded peripherals.
- Designed 2 custom PCBs and developed additional prototypes using breadboard-based hardware integration.
- Managed the complete development lifecycle, including requirements analysis, component selection, firmware/software development, hardware integration, debugging, testing, and delivery.
- Generated approximately ₹1.5 lakh in revenue through independent engineering development and delivery.

PROJECTS

SKYCLEAN: A LOW COST WINDOW CLEANING DEVICE *May 2025*

- Designed a low-cost window-cleaning robot using BLDC-based suction and wheel-driven vertical navigation.
- Developed a manual control interface for real-time control and adjustment of robot movement.
- Designed and integrated electromechanical subsystems for stable vertical operation.
- Ranked Top 5 among 4,000 teams at the TECHgium 8th Edition conducted by LTTS.

RFID BASED ECOSYSTEM FOR ACADEMIC INSTITUTIONS *Feb 2025*

- Developed an RFID-based campus ecosystem that extended existing student ID cards into a unified platform for identity, attendance, library automation, and payments.
- Built a database-driven self-service library system where students could issue and return books by scanning their RFID ID card and book barcode, reducing manual intervention.
- Extended the platform for bus, laboratory, and classroom attendance, using the same RFID identity infrastructure across multiple campus operations.
- Designed an RFID-enabled campus food-court wallet, integrating student balance management with Razorpay for online top-ups and RFID-based payments at the point of purchase.

FAULT DETECTION IN MICROCONTROLLERS *May 2024*

- Developed a platform for identifying physical and logical faults in microcontroller-based systems.
- Built a hardware prototype using ESP32-CAM and Arduino development platforms.
- Implemented mechanisms for detecting and evaluating microcontroller-level failures.
- Explored soft real-time system behavior and embedded fault detection techniques.

BUCKLE UP FOR A SAFER RIDE: CAR SAFETY SYSTEM *Jun 2023*

- Developed a microcontroller-based safety interlock system linking seatbelt status with vehicle starter operation.
- Implemented firmware on a PIC microcontroller for real-time monitoring and control.
- Implemented CAN communication for reliable exchange of vehicle safety data.
- Integrated embedded control logic with the vehicle-starting mechanism to enforce safety conditions.

SKILLS

Programming	: C, Embedded C, C++, Python, Verilog
Embedded Systems	: Real-Time Systems, Hardware-Software Integration, Embedded Debugging
Communication Protocols	: CAN, UART, SPI, I²C
Hardware	: ESP32, ESP8266, PIC, Arduino
Development Tools	: Keil µVision, Arduino IDE, Proteus, MATLAB
Engineering	: Circuit Analysis, Microcontroller Programming, System Integration, Software Testing

EDUCATION

SRI RAMAKRISHNA ENGINEERING COLLEGE — *Coimbatore, TamilNadu*
Bachelor of Engineering in Electronics and Communication Engineering — *Nov 2022 - May 2026*
Cumulative GPA: 7.69

VIDHYA NIKETAN MATRICULATION SCHOOL — *Coimbatore, TamilNadu*
HSC — *Jul 2020 - Jul 2022*
Percentage: 86%

VIDHYA NIKETAN MATRICULATION SCHOOL — *Coimbatore, TamilNadu*
SSLC — *Jul 2020*
Percentage: 79.2%

CERTIFICATIONS

Introduction to Embedded System Design — NPTEL
Introduction to Industry 4.0 and Industrial Internet of Things — NPTEL
Foundations of Cybersecurity — Google

ACHIEVEMENTS & LEADERSHIP

- Participated and placed Top 5 in SAFETYTHON – A Road Safety Hackathon conducted by Kumaraguru College of Technology, Coimbatore on 19th April 2023
- Finalist of TECHgium 8th Edition conducted by L&T Technology Services.
- Joint Secretary of Fine Arts Club of Sri Ramakrishna Engineering College for the year 2024 – 2025.
- Executive Member in the Social Innovation Club (SIC) for the year 2024 – 2025.
- Placed 1st in the Paper Presentation conducted by Department of Science and Humanities, SREC.